Blinkit Grocery Sales Analysis

End-to-End Data Analyst Portfolio Project

1. Executive Summary

This project analyzes grocery sales performance across products, fat-content segments, outlet types, outlet sizes, location tiers, pricing, visibility, and customer ratings. The objective is to identify the major sales drivers, compare outlet performance, surface high-performing product categories, and convert the findings into actionable business recommendations. The project combines Excel data preparation, Python exploratory analysis, MySQL analysis, and an interactive Power BI dashboard.

2. Business Problem

- Measure overall sales performance using clear business KPIs.
- Identify the product categories and segments contributing the most revenue.
- Compare outlet performance by type, size, and location tier.
- Understand relationships between sales, product visibility, price, and ratings.
- Provide recommendations for inventory planning and outlet strategy.

3. Dataset Overview

Area	Fields / Examples
Product	Item Identifier, Item Type, Item Weight, Fat Content
Pricing	Item MRP
Visibility	Item Visibility
Outlet	Outlet Identifier, Outlet Type, Outlet Size
Location	Outlet Location Type
Outlet Age	Outlet Establishment Year
Performance	Sales
Customer feedback	Rating

4. Data Cleaning

- Standardized categorical values such as Low Fat and Regular.
- Converted numeric fields to appropriate numeric data types.
- Checked duplicate records and missing values.
- Handled missing item-weight values using median imputation by item type where possible.
- Validated the final dataset before analysis.

5. Python Analysis

The Python script uses Pandas and Matplotlib to load the Excel dataset, normalize column names, clean the data, calculate KPIs, aggregate sales by product and outlet dimensions, generate top-item analysis, export data-quality tables, and create portfolio-ready charts. The script is designed to work with common variations of the Blinkit dataset column names.

6. SQL Analysis

- Overall KPI calculations and data-quality checks.
- Revenue by item type and fat content.
- Revenue by outlet type, outlet size, location tier, and establishment year.
- MRP, item visibility, and rating analysis.
- Top-product and outlet-level performance.
- Window-function ranking and revenue-share analysis.
- Business-question queries for management decision support.

7. Power BI Dashboard

The Power BI component is structured as an executive KPI dashboard with supporting analysis pages. The project plan uses four analysis areas: Executive Summary, Product Analysis, Outlet Analysis, and Customer/Insight Analysis. The dashboard uses DAX measures and interactive filtering to make sales, product, outlet, and rating insights easy to explore.

Recommended KPI measures

- Total Sales = SUM(Sales)
- Average Sales = AVERAGE(Sales)
- Total Items = COUNT(Item Identifier)
- Average Rating = AVERAGE(Rating)
- Total Outlets = DISTINCTCOUNT(Outlet Identifier)

8. Key Analytical Questions

Question	Analysis
Which product categories lead revenue?	Group sales by Item Type and rank descending.
Which outlet type performs best?	Compare revenue and average sales by Outlet Type.
Which location tier contributes most?	Aggregate revenue by Outlet Location Type.
Does fat content affect sales?	Compare revenue and average sales by Fat Content.
Which products are strongest?	Rank individual items by Sales.
Where are inventory opportunities?	Combine category, visibility, sales, and outlet analysis.

9. Business Recommendations

- Prioritize inventory and promotions around consistently high-revenue categories.
- Use outlet-level rankings to identify strong formats and locations for expansion or replication.
- Monitor low-performing categories and outlets for assortment, visibility, and pricing opportunities.
- Use visibility and sales together to identify products that may require better placement or promotion.
- Track customer ratings alongside revenue so growth initiatives do not reduce customer satisfaction.
- Refresh the Power BI dashboard periodically so management can monitor KPI movement and outlet performance.

10. Tools & Skills Demonstrated

Excel • Python • Pandas • NumPy • Matplotlib • MySQL • SQL • Power BI • DAX • Power Query • Data Cleaning • EDA • KPI Reporting • Data Visualization • Business Insights

11. GitHub Deliverables

- Blinkit_Grocery_Sales_Complete_SQL.sql
- Blinkit_Grocery_Sales_Analysis.py
- Blinkit Power BI .pbix file
- Dashboard screenshots
- Source Excel dataset
- This project report
- README.md

12. Conclusion

This project demonstrates an end-to-end analytics workflow: raw grocery data is cleaned and validated, explored with Python, queried with SQL, and communicated through Power BI. The final portfolio package is designed to demonstrate practical Data Analyst skills rather than only dashboard creation, with emphasis on business questions, KPI reporting, segmentation, ranking, and actionable recommendations.